Boundary CFT$_3$-on-S^3 observables and the GeoVac spectral triple: partition function and wedge KMS entropy

J. Loutey*

May 2026 (preprint)

Abstract

We exhibit three complementary boundary-side quantities, framed by standard CFT$_3$-on-S^3 calculations, around the GeoVac spectral triple, each obtained by a structurally distinct operation — (A) from the full, untruncated continuum spectra of the conformally coupled scalar and Camporesi–Higuchi Dirac operators on the round S^3, (B) from the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ built on the truncated triple from a posited wedge generator, (C) by comparing that generator with a continuum one: *(A)* the universal F-theorem coefficient F_{S^3}, the standard zeta-regularised determinant $-\frac{1}{2}\zeta'_\Delta(0)$ of the continuum spectrum, is re-derived in closed Hurwitz form and agrees with the Klebanov–Pufu–Safdi (2011) values (symbolically for the Dirac, to 61 digits with PSLQ identification for the scalar) for both the conformally coupled scalar ($F_s = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2}$) and the free Camporesi–Higuchi Dirac ($F_D = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2}$) — an independent re-derivation of known continuum values, in which the finite cutoff $n_{\max}$ never enters; *(B)* the wedge KMS entropy on the truncated triple has the exact asymptotic $S(\rho_W) = 2\log n_{\max} + (\coth 1 - \log(2\sinh 1)) + O(1/n_{\max})$, whose slope 2 is the exponent of the $\sim n_{\max}^2$ degeneracy of the lowest shell of the posited generator K_α^W (the slope is independent of the inverse temperature; the constant is the unit-β value, and ρ_W is not the Bisognano–Wichmann-normalised state, which would be $e^{-\pi K_\alpha^W}/Z$ since K_α^W carries the values of $2J_z$; the 2 is not the dimension of the entangling surface, which for a CFT$_3$ is a circle); *(C)* the framework's wedge generator K_α^W (a posited relabelling built from the polar rotation generator $J_{\text{polar}} = 2J_z$, not derived from a state) shares a discrete integer spectrum and closure at rotation angle 2π (its adjoint flow closes at $t = \pi$) with the *Euclidean* rotation generator whose Wick rotation is the continuum Casini–Huerta–Myers modular Hamiltonian of a hemispherical cap of the spatial S^2 (the latter itself generates a boost-like flow with continuous spectrum); this is a structural *analogy* with the Euclidean generator only, and the identification of K_α^W as the modular Hamiltonian of a wedge-restricted state is **[OPEN]** (2026-09-28; see [12]). The scalar/Dirac free energies decompose orthogonally into their two transcendental sectors: $F_D + 2F_s = \log(2)/2$ isolates the $\log 2$ sector (a level-2 cyclotomic period, in the ring of the M3 sub-mechanism of the master Mellin engine of [2]), and $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ isolates the $\zeta(3)/\pi^2$ sector (an M1×M3 cross-product), following the engine's period-ring stratification [2]; no Seeley–DeWitt (M2) content appears in these values. This is a decomposition of the continuum KPS values, and the first application of the master Mellin engine's tagging to independently published (non-GeoVac-internal) values. The F-theorem values themselves are due to [24] and are not new; the present paper's contributions are (i) their independent re-derivation from the continuum spectra via closed Hurwitz forms (with PSLQ confirmation), (ii) the orthogonal $\log 2$ / $\zeta(3)/\pi^2$ dual-basis decomposition of those continuum values, classified as M3-ring and M1×M3 content (Theorem 3.3), and (iii) the exact large-$n_{\max}$ asymptotic of the wedge-KMS entropy on the

*Independent researcher. jloutey@gmail.com. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

truncated triple, a lowest-shell degeneracy count. The framework has no bulk, so none of this is holographic. The CHM comparison (C) is recorded as an analogy only: K_α^W is posited, and a GH distance between spectral triples does not bound the convergence of one operator to another. The bulk-side identifications (Ryu–Takayanagi minimum surfaces, JLMS bulk-modular reconstruction, AdS_4 / H^4 dual geometry) are blocked at the framework's existing infrastructure: zero AdS / H^4 machinery, and the Wick-rotation $S^3 \to H^3$ supplies no natural discrete subgroup [6].

1 Introduction

The GeoVac spectral triple [1, 7] is a finite-cutoff operator-system truncation of the Camporesi–Higuchi spectral triple [20] on the round S^3. Its modular structure on the hemispheric wedge has been developed across Papers 42 and 44–49 [12–18] (the Krein-carrier extension, formerly Paper 43, is now Paper 42 § 11): with the posited geometric BW-α generator K_α^W (the diagonal wedge relabelling built from J_{polar}; integer spectrum) and the Tomita–Takesaki BW-γ modular operator of the Gibbs state built from it, the flow closes, $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$, bit-exactly — automatically, for any integer-spectrum generator, so this is not a finite-cutoff identification of the four-witness Wick-rotation theorem (deriving the generator from a Camporesi–Higuchi wedge state is open, Paper 42). The Lorentzian *propinquity* extension to signature $(3, 1)$ is descoped: Paper 45 [14] proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically, closing that route) and the strong-form construction [15] is correspondingly descoped on the natural substrate, so a Lorentzian quantum-metric convergence remains an open target. What survives is the norm-resolvent non-compact-carrier closure [16] at the spectral level and Paper 45's signature-agnostic product-carrier rate. The Krein–Mondino–Sämann bridge [17] and the OSLPLS strong-form bridge [18] proposed a synthetic-Lorentzian geometric interpretation; both are descoped and archived (see the closing section).

The present paper turns from the math.OA structural content of the modular machinery to its *physics-side observables*. We ask what the framework's machinery yields on boundary observables corresponding to standard CFT_3-on-S^3 calculations, and whether the resulting structural pattern exhibits any genuinely new content beyond a numerical match. Two different objects enter, and we keep them apart. The partition-function side (A) uses the full continuum spectra of the scalar and Dirac operators — the spectra whose truncations form the GeoVac triple, but not the truncations themselves. The wedge-state side (B, C) lives on the truncated triple at finite $n_{\max}$.

Headline results. We establish three complementary boundary-side observables and one structural decomposition theorem:

1. **Partition function re-derivation** (Sec. 3): the universal F-theorem coefficient, evaluated as the zeta-regularised determinant of the full continuum spectrum in closed Hurwitz form, agrees with the Klebanov–Pufu–Safdi (2011) [24] values for both the conformally coupled scalar (to 61 digits, PSLQ-identified) and the free Camporesi–Higuchi Dirac (symbolically) on the round S^3. This is an independent re-derivation of known values, not a finite-cutoff result: the truncation $n_{\max}$ never enters.

2. **Master Mellin engine decomposition** (Theorem 3.3): the scalar and Dirac partition functions decompose into linear combinations of $\log 2$ and $\zeta(3)/\pi^2$, and a dual-basis pair of combinations isolates each sector orthogonally; in the period-ring stratification of the master Mellin engine [2] the first is an M3-ring (level-2 cyclotomic) period and the second an M1×M3 cross-product. This is a decomposition of the continuum KPS values, and the first application of the engine's classification to independently published (non-GeoVac-internal) values.

3. **Wedge KMS entropy structural classification** (Sec. 4): the von Neumann entropy of ρ_W scales as $S(\rho_W) \sim 2 \log n_{\max}$; the slope 2 is the exponent of the $n_{\max}(n_{\max}+1) \sim n_{\max}^2$ degeneracy of the lowest shell of the posited generator K_α^W **[MEASURED]** (degeneracy count, checked against direct diagonalisation) — a property of the truncation, not a continuum geometric dimension. The F-coefficient does *not* appear in $S(\rho_W)$: the continuum spectral-zeta side and the finite-cutoff wedge-state side encode *different* quantities.

4. **Casini–Huerta–Myers analogy** (Sec. 5): the framework's posited wedge generator K_α^W shares integer spectrum and 2π-periodic flow with the Euclidean rotation generator about the entangling circle, whose Wick rotation is the continuum CHM modular Hamiltonian of a hemispherical cap H_+ of the spatial S^2, $K_{\text{cont}} = 2\pi \int_{H_+} \xi \cdot T^{00}\, dA$ [32]; K_{cont} itself generates a boost-like flow with continuous spectrum, so the analogy is with the Euclidean generator only. **[OPEN]**: identifying K_α^W as a modular Hamiltonian (deriving it from a wedge-restricted state) and any operator-level convergence $K_\alpha^W \to K_{\text{cont}}$; the Paper 38 GH bound concerns the spectral triples, not this operator.

Bulk-side scope. Bulk-side identifications — Ryu–Takayanagi minimum-surface entanglement entropy [35], the JLMS [36] bulk-modular reconstruction, and AdS_4 / H^4 dual geometry — are blocked at the framework's existing infrastructure: zero AdS_4 / H^4 machinery exists, and a previous attempt at the natural Wick continuation $S^3 \to H^3$ [6] found no discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants. We document this scope statement honestly in Sec. 6, with no claim that the bulk side is reachable from the framework's current axiom set.

2 Setup

2.1 The GeoVac spectral triple on S^3

The Fock projection [1, 19] maps the single-particle hydrogen Hamiltonian conformally to the Laplace–Beltrami operator on S^3 via stereographic projection in momentum space. The framework's spectra on the unit S^3 are exact in $\mathbb{Q}$:

- **Scalar Laplacian.** Eigenvalues $\lambda_n^2 = n(n+2)$ with multiplicity $d_n = (n+1)^2$ for $n = 0, 1, 2, \ldots$. After the conformal mass shift $\xi R_{\text{scalar}} = \frac{1}{8} \cdot 6 = \frac{3}{4}$ for the conformally coupled scalar, the spectrum becomes $(n + \frac{1}{2})(n + \frac{3}{2})$ with the same multiplicity.
- **Camporesi–Higuchi Dirac.** Eigenvalues $|\lambda_n| = n + \frac{3}{2}$ with multiplicity $g_n^{\text{Dirac}} = 2(n+1)(n+2)$ for the single (3d) Dirac fermion (the standard S^3 massless Dirac, whose F_D is re-derived below from the continuum spectrum; in odd dimension there is no Weyl projection), $4(n+1)(n+2)$ for the doubled four-component Dirac. Both carry the same Hurwitz-zeta structure under analytic continuation [4, 5].

The truncation at level $n_{\max}$ retains all $n \le n_{\max}$ shells and defines the operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}} \mathcal{A} P_{n_{\max}}$ in the Connes–van Suijlekom framework [21]. Convergence $\mathcal{O}_{n_{\max}} \to \mathcal{O}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance holds with the Fejér-kernel bound $\gamma_{n_{\max}} \sim (4/\pi) \log(n_{\max})/n_{\max}$ [9] (a tapered amplitude gives $O(1/n_{\max})$ **[MEASURED]**); the constant $4/\pi$ is stated in the corpus's dual-Coxeter normalisation ($\text{Cas}(\text{ad}) = h^\vee$, under which the $SU(2)$ geodesic distance is the rotation angle) and is convention-dependent, not canonical: on the unit S^3 the same bound reads $2/\pi = \text{Vol}(S^2)/\text{Vol}(S^3)$, the volume quotient that carries its M1 content in the master Mellin engine [2]. The partition-function values of Sec. 3 use the untruncated ($n = 0, 1, 2, \ldots$) spectra and do not depend on this truncation; the truncation enters only the wedge-state side (Secs. 4–5).

2.2 Wedge KMS state

The hemispheric wedge W is $P_W = \frac{1}{2}(I + R_{\text{polar}})$, the $+1$ eigenspace of the reflection $m_j \mapsto -m_j$ ([12]), whose basis states are labelled by their $m_j > 0$ component. Per [12] Def 5.1, the posited BW-α wedge generator (its identification as a modular Hamiltonian is open) is

$$K_\alpha^W = \text{diag}(2m_j) \text{ on the wedge states, built from } J_{\text{polar}} = 2J_z,$$

where J_{polar} is the polar rotation generator, used on the wedge through the diagonal relabelling of Paper 42 Def 5.1 (it does not preserve the wedge); K_α^W has integer eigenvalues $2m_j \in \{1, 3, 5, \ldots, 2n_{\max} - 1\}$ and multiplicity $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$ for $k = 0, \ldots, n_{\max} - 1$. The wedge state used throughout is the unit-β Gibbs state of K_α^W (not the Bisognano–Wichmann-normalised state; see Remark 5.1)

$$\rho_W = \frac{e^{-K_\alpha^W}}{Z}, \qquad Z = \text{Tr}_{\mathcal{H}_W}\big(e^{-K_\alpha^W}\big).$$

This is the substrate for the wedge-side observables (B) and (C); observable (A) uses the continuum spectra only.

3 Partition function: closed-form re-derivation of the KPS values

3.1 Continuum reference

For the round unit S^3, the free energies of free CFTs computed via Hurwitz-zeta regularization are [24]:

$$F_s^{\text{KPS}} = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} \approx 0.063807, \tag{1}$$

$$F_D^{\text{KPS}} = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} \approx 0.218960, \tag{2}$$

for the conformally coupled scalar and the free massless Dirac fermion (no Weyl projection in odd dimension), respectively. Both values are expressed in terms of the functional determinant identities $F = -\frac{1}{2}\zeta'_\Delta(0)$ (scalar) and $F = \zeta'_{|D|}(0)$ (Dirac), with the analytic continuation of the divergent spectral sum supplied by the standard zeta-function regularization. The transcendental signature involves only $\log 2$, $\zeta(3)$, and π^2.

3.2 Closed Hurwitz forms of the continuum spectral zeta

The spectral zeta function of the full (untruncated, $n = 0, 1, \ldots, \infty$) continuum spectrum on S^3 admits a closed-form Hurwitz expansion. No finite cutoff enters: the sums run over all n, exactly as in [24]; what is specific here is only the closed Hurwitz form, the half-integer identity used throughout the framework's QED modules. For the Dirac case, the Dirichlet series can be written exactly via the half-integer Hurwitz identity $\zeta_H(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$:

$$D_{\text{Dirac}}(s) = \sum_{n=0}^{\infty} \frac{g_n^{\text{Dirac}}}{|\lambda_n|^s} = 2\big(2^{s-2} - 1\big)\zeta_R(s-2) - \tfrac{1}{2}\big(2^s - 1\big)\zeta_R(s). \tag{3}$$

This identity is encoded in the production module `geovac/qed_two_loop.py` and is verified symbolically across integer s.

For the conformally coupled scalar, the spectral zeta admits the Hurwitz-asymptotic expansion

$$\zeta_\Delta(s) = \sum_{n=0}^{\infty} \frac{(n+1)^2}{[(n+1/2)(n+3/2)]^s} = \sum_{k \geq 0} g_k(s)\, \zeta_R(2s + 2k - 2), \tag{4}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ the binomial-like coefficient.

3.3 Closed-form match: scalar

Differentiating (4) at $s = 0$ and using $g_k(0) = \delta_{k,0}$, $g_k'(0) = 1/(k \cdot 4^k)$ for $k \geq 1$, plus the standard identities $\zeta_R'(-2) = -\zeta(3)/(4\pi^2)$, $\zeta_R(0) = -\frac{1}{2}$, $\zeta_R(2k-2)$ expressible in $\pi^{2k-2} \cdot \mathbb{Q}$ for $k \geq 2$:

$$\zeta_\Delta'(0) = 2\zeta_R'(-2) + \sum_{k \geq 1} \frac{\zeta_R(2k-2)}{k \cdot 4^k}. \tag{5}$$

Theorem 3.1 (Closed-form scalar partition function (numerical + PSLQ)). ***[NUMERICAL, 61 digits + PSLQ; value prior art [24]]*** *The zeta-derivative of the continuum conformal-scalar spectrum (4) satisfies*

$$F_s := -\tfrac{1}{2}\,\zeta_\Delta'(0) = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} = F_s^{KPS}, \tag{6}$$

in agreement with (1) to 61 digits, with the closed form identified by PSLQ.

Proof sketch. The asymptotic Hurwitz expansion (5) converges geometrically in k at rate $1/4$ per term; truncation at $k_{\max} = 100$ gives 61+ matching digits at dps = 200 when compared to $-2F_s^{\mathrm{KPS}}$ via `mpmath`. PSLQ at coefficient bound 10^6 and tolerance 10^{-40} finds the integer relation $[8, 2, -3]$:

$$8 \cdot \zeta_\Delta'(0) + 2\log 2 - 3 \cdot \frac{\zeta(3)}{\pi^2} = 0,$$

which is exactly the claimed identity. See driver `debug/ads_track_a_scalar_partition_function.py` and JSON `debug/data/ads_track_a_scalar_partition_function.json` for full computation. □

3.4 Closed-form match: Dirac

Differentiating (3) symbolically at $s = 0$, using the same Riemann-zeta identities at $s = -2, 0$:

$$D_{\mathrm{Dirac}}'(0) = \tfrac{1}{2}\log 2 \cdot \zeta_R(-2) + \left(-\tfrac{3}{2}\right)\zeta_R'(-2) - \tfrac{1}{2}\left[\log 2 \cdot \zeta_R(0) + 0\right] \tag{7}$$

$$= 0 - \tfrac{3}{2} \cdot \left(-\tfrac{\zeta(3)}{4\pi^2}\right) - \tfrac{1}{2} \cdot \left(-\tfrac{\log 2}{2}\right) = \frac{3\zeta(3)}{8\pi^2} + \frac{\log 2}{4}. \tag{8}$$

Theorem 3.2 (Closed-form Dirac partition function (symbolic)). ***[SYMBOLIC; value prior art [24]]*** *The zeta-derivative of the continuum Camporesi–Higuchi Dirac spectrum (3) satisfies, as an exact symbolic identity,*

$$F_D := D_{\mathrm{Dirac}}'(0) = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} = F_D^{KPS}, \tag{9}$$

agreeing symbolically with (2).

Proof. Direct symbolic computation; `sympy.simplify(D'(0) - F_D)` returns 0. See driver `debug/ads_track_a_dirac_partition_function.py`. □

3.5 Master Mellin engine decomposition

The master Mellin engine of [2] §III.7 tags (a census and a reading) the transcendental signatures of GeoVac observables via the three sub-mechanisms

$$\mathcal{M}[\mathrm{Tr}(|D|^k \cdot e^{-tD^2})], \qquad k \in \{0, 1, 2\},$$

(the absolute trace: the signed $k = 1$ trace vanishes on the symmetric Camporesi–Higuchi spectrum), labeled M1 ($k = 0$), M3 ($k = 1$), and M2 ($k = 2$). We follow the period-ring

stratification of [2] §III.7 (the single owner of the slot assignments and ring homes, with Paper 55 as the census; restated here only to the extent this section uses it): M1 (the $k = 0$ volume slot) lies in $\mathbb{Q}[\pi, \pi^{-1}]$; M2 (Seeley–DeWitt heat-kernel coefficients) lies, at integer s, in the pure-Tate ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, with no $\log 2$ and no $\zeta(3)$ (at half-integer s the Hurwitz functions $\zeta(2s, 3/2)$ and $\zeta(2s-2, 3/2)$ of the spinor spectrum sit at odd-integer argument and yield odd zeta values, which are M3's: a value reached from two slots is placed by the period ring it lands in, and these lie outside M2's ring); M3 (quarter-integer Hurwitz / vertex-parity sums) lies in the cyclotomic mixed-Tate periods $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4, a ring which contains the odd zeta values and the level-2 cyclotomic period $\log 2$.

Theorems 3.1 and 3.2 exhibit F_s and F_D as linear combinations of $\log 2$ and $\zeta(3)/\pi^2$. In the Dirac computation, $\log 2$ enters through the 2^s factor of the half-integer Hurwitz identity $\zeta_H(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$ (eq. 3); it is a level-2 cyclotomic period and so lies in M3's ring. $\zeta(3)/\pi^2$ enters through $\zeta_R'(-2) = -\zeta(3)/(4\pi^2)$; $\zeta(3)$ lies in M3's ring and $1/\pi^2$ in M1's, so it is an M1×M3 cross-product (the labelling of [3]). Neither summand carries Seeley–DeWitt (M2) content. An earlier version of this paper labelled $\log 2$ as a Seeley–DeWitt (M2) transcendental, via $\zeta_R'(0)$, and read the two combinations below as an M2/M3 split; that labelling is withdrawn (2026-09-29) in favour of the stratification of [2].

Theorem 3.3 (Scalar/Dirac $\log 2$ / $\zeta(3)/\pi^2$ orthogonal decomposition)**.** ***[SYMBOLIC]*** *The continuum free energies F_s and F_D project orthogonally onto the $\log 2$ sector (an M3-ring, level-2 cyclotomic period) and the $\zeta(3)/\pi^2$ sector (an M1×M3 cross-product) of the master Mellin engine's period-ring stratification [2] via the dual-basis combinations*

$$F_D + 2\,F_s = \frac{\log 2}{2} \qquad \textit{(isolates the } \log 2 \textit{ sector; all } \zeta(3) \textit{ cancels),} \tag{10}$$

$$F_D - 2\,F_s = \frac{3\zeta(3)}{4\pi^2} \qquad \textit{(isolates the } \zeta(3)/\pi^2 \textit{ sector; all } \log 2 \textit{ cancels).} \tag{11}$$

Proof. Substituting (1) and (2):

$$F_D + 2F_s = \left(\tfrac{\log 2}{4} + \tfrac{3\zeta(3)}{8\pi^2}\right) + 2\left(\tfrac{\log 2}{8} - \tfrac{3\zeta(3)}{16\pi^2}\right) = \tfrac{\log 2}{4} + \tfrac{\log 2}{4} = \tfrac{\log 2}{2},$$

and similarly $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$. The two combinations span a 2-dimensional space; the projection coefficients $(1, +2)$ and $(1, -2)$ act as a dual basis for the $\log 2$ / $\zeta(3)/\pi^2$ decomposition on the (scalar, Dirac) plane. □

Remark 3.4 (First application to non-GeoVac-internal data)**.** Theorem 3.3 is the first application of the master Mellin engine's tagging to values that are *not* GeoVac-internal. Previous verifications — the case-exhaustion theorem of [7] §VIII, the 208/208 Prediction 1 panel of [8], the modular-residual closed form of Sprint MR-B[1] — all operated on framework-internal quantities. The KPS partition functions (1), (2) are independently-published continuum CFT_3 data, and the engine's stratification [2] tags their transcendental content as M3-ring ($\log 2$, from the half-integer Hurwitz 2^s factor) and M1×M3 ($\zeta(3)/\pi^2$, from $\zeta_R'(-2)$) — not as Seeley–DeWitt (M2) content. The content is a decomposition of continuum values: the numbers are KPS's, re-derived here from the same continuum spectra in closed Hurwitz form — not produced by the finite-cutoff triple, and not an independent prediction of CFT physics.

4 Wedge KMS entropy: state-side observable

4.1 The framework's wedge KMS entropy

The von Neumann entropy of the wedge state $\rho_W = e^{-K_\alpha^W}/Z$ (unit inverse temperature for K_α^W; not the Bisognano–Wichmann normalisation, Remark 5.1) is computed directly by di-

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

agonalizing K_α^W and applying $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ (recomputed from the live BW construction, the slope-2 scaling, the $\coth 1 - \log(2\sinh 1)$ constant, and the area-law rejection, in `tests/test_paper50_wedge_kms.py`). Sprint BH-Phase0 (2026-05-22) [38] first reported a small-cutoff log-linear fit $S(\rho_W) \approx 1.94 \log n_{\max} + 0.56$ over $n_{\max} \in \{2, \ldots, 7\}$. A subsequent reanalysis (forcing-catalogue Door 2, 2026-06-01, `debug/door2_bw_entropy_reread_memo.md`) replaced this fit by the exact $n_{\max} \to \infty$ resummation of the closed-form entropy $S = \log Z + \langle K_\alpha^W \rangle$ (which reproduces the direct diagonalization to $\leq 4 \times 10^{-16}$). The leading $n_{\max}^2$ shell degeneracy $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$ forces the slope to be *exactly* 2, and the additive constant has the exact closed form

$$S(\rho_W) = 2 \log n_{\max} + \big(\coth 1 - \log(2 \sinh 1)\big) + O(1/n_{\max}), \qquad \coth 1 - \log(2\sinh 1) = 0.458448\ldots, \tag{12}$$

the constant verified by PSLQ to residual $< 10^{-80}$ (integer relation $[-1, 1, -1, -1]$ against $\{C_\infty, \coth 1, \log 2, \log \sinh 1\}$). The windowed least-squares slope climbs monotonically $1.94 \to 2.00005$ as the fit window moves to $n_{\max} \in [350, 2000]$; the original 1.94 is a small-$n_{\max}$ artifact of a sequence converging to exactly 2. The slope 2 is the exponent of the lowest-shell degeneracy $n_{\mathrm{eq}} = n_{\max}(n_{\max} + 1)$ of the posited generator K_α^W (mechanism below) **[MEASURED]**; an earlier version read it as the dimension of the entangling surface, which is withdrawn (2026-09-28): for a CFT_3 the entangling surface of a hemispherical cap is a circle S^1, not a two-sphere. The additive constant lives in the Boltzmann-base ring (functions of e via $\coth 1$, $\sinh 1$, tied to the integer spacing $\Delta K_\alpha^W = 2$ at unit β); it carries no master Mellin engine transcendental, consistent with Prop. 4.1. A PSLQ search against the F-theorem ring $\{1, \log 2, \zeta(3)/\pi^2\}$ at coefficient ceiling 10^8 returns null: the wedge-entropy constant and the F-coefficient show no integer relation (numerically, not a proof of disjoint rings), sharpening Prop. 4.1 from "F absent from $S(\rho_W)$" to "the state-side constant admits no integer relation with the spectral-zeta ring up to coefficient ceiling 10^8" (a strong numerical disjointness, not a proof of ring non-membership). The structural support is separate for the two terms: the leading $2 \log n_{\max}$ is the logarithm of an integer degeneracy, hence ring-free; the additive constant is *not* the logarithm of an integer, but a closed-form function of e alone ($\coth 1$, $\sinh 1$), i.e. in the Boltzmann-base ring by construction.

4.2 Mechanism: degeneracy log on the lowest K_α^W shell

The K_α^W spectrum is bounded above by $2n_{\max} - 1$ at finite cutoff. In ρ_W (unit β), the per-state weight at $K_\alpha^W = 1$ (the lowest shell) is e^{-1}/Z, while at $K_\alpha^W = 3$ it drops to e^{-3}/Z, a factor of $e^{-2} \approx 0.135$ suppression per shell. The partition function is therefore dominated by the lowest shell of multiplicity $n_{\mathrm{eq}} = n_{\max}(n_{\max} + 1) \sim n_{\max}^2$:

$$Z \approx n_{\mathrm{eq}} \cdot e^{-1} \approx 0.368\, n_{\max}^2,$$

giving effective probability $p_{\mathrm{eq}} \approx 1/n_{\mathrm{eq}}$ per lowest-shell state, and hence

$$S(\rho_W) \approx -n_{\mathrm{eq}} \cdot p_{\mathrm{eq}} \log p_{\mathrm{eq}} = \log n_{\mathrm{eq}} \approx 2 \log n_{\max}.$$

The two asymptotic limits in s-deformation $\rho(s) = e^{-sK_\alpha^W}/Z$, $s \to 0^+$ (maximally mixed on $\mathcal{H}_W$, giving $S \to \log \dim \mathcal{H}_W$) and $s \to \infty$ (maximally mixed on the lowest shell, giving $S \to \log n_{\mathrm{eq}}$), are saturated bit-exactly [38].

4.3 Structural decomposition: F is in the spectral-zeta side, not the state side

Proposition 4.1 (F vs $S(\rho_W)$ live in different sectors)**.** *Two structurally distinct calculational operations extract different quantities:*

- ***Spectral-zeta side:*** *the analytic continuation* $-\frac{1}{2}\zeta'_\Delta(0)$ *of the continuum spectrum extracts the universal F-theorem coefficient* F_s^{KPS} *via Theorem 3.1; similarly for* F_D^{KPS} *via Theorem 3.2. The transcendental signature is in the ring* $\mathbb{Q}\cdot\log 2 \oplus \mathbb{Q}\cdot\zeta(3)/\pi^2$ *(M3-ring and M1×M3 in the master Mellin engine's stratification [2]).*

- ***Wedge KMS state side:*** *the von Neumann entropy* $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ *extracts the lowest-shell degeneracy of the posited generator* K_α^W*, with leading scaling* $2\log n_{\max}$ *(the degeneracy grows as* $n_{\max}^2$*). The leading term is ring-free (a logarithm of an integer).*

The two operations act on different objects (the first on the continuum spectra, whose truncations form the GeoVac triple; the second on the wedge state ρ_W *built on the truncated triple at finite* $n_{\max}$*) and extract* categorically different *quantities: the universal F-coefficient versus a lowest-shell degeneracy count. The F-coefficient does* not *appear in* $S(\rho_W)$*.*

This decomposition is consistent with the PSLQ-null finding (no detected integer relation; not a proof of disjointness) of Sprint TD Track 5 (2026-05-09) [39]: no integer relation was detected between GeoVac entropies and the master Mellin engine ring, while spectral-side observables are read inside it.

4.4 Three blockers for literal Ryu–Takayanagi on the framework's graph

The continuum Ryu–Takayanagi formula [35] identifies the boundary entanglement entropy of a region A with the area of the bulk minimum surface anchored at ∂A, weighted by $1/(4G_N)$. A literal discrete analog on the framework's graph would require: (i) a well-defined notion of graph minimum-cut between spatial sub-regions; (ii) a perfect-tensor structure at graph vertices ensuring isometric bulk-boundary encoding; (iii) a bulk dual geometry.

Proposition 4.2 (Three blockers for literal RT on the framework)**.** *The framework's discrete* S^3 *graph at finite* $n_{\max}$ *exhibits three independent structural obstructions to a literal RT formula:*

1. *The Fock graph at level* $n_{\max}$ *has* $\beta_0 = n_{\max}$ *connected components (one per* ℓ*-sector), so the naive graph-cut between spatial sub-regions is identically zero or trivially confined to a single* ℓ*-sector.*

2. *The Wigner 3j symbols at graph vertices are Clebsch–Gordan projections rather than isometries; the HaPPY tensor-network RT proof [37] requires perfect tensors at network nodes.*

3. *No bulk-dual construction exists in the framework's infrastructure. The natural candidate — Wick continuation* $S^3 \to H^3$ *— was closed by Sprint RH-B (2026-04-17) [6] as a clean structural dead end: no discrete subgroup* $\Gamma \subset SO(3,1)$ *is selectable from framework invariants.*

These three blockers are not in tension with the boundary-side results of Sec. 3 or Sec. 5: the F-coefficients of Sec. 3 are computed from the continuum operator spectra (not from the wedge state), the wedge entropy is a lowest-shell degeneracy count, and the CHM comparison is an analogy; none of these requires, and the framework cannot independently construct, a bulk dual against which to compare via RT.

5 Casini–Huerta–Myers analogy

5.1 Continuum CHM hemisphere modular Hamiltonian

For a CFT_3 the Lorentzian spatial slice is a two-sphere, and the Casini–Huerta–Myers construction [32] (valid for any CFT, not only free ones) gives the modular Hamiltonian of the vacuum

restricted to a hemispherical cap $H_+ \subset S^2$ as

$$K_{\text{cont}} = 2\pi \int_{H_+} \xi(x) \cdot T^{00}(x)\, dA(x), \tag{13}$$

where ξ is the conformal Killing vector field preserving the causal development of the cap (vanishing on the entangling surface $\partial H_+ = S^1$, the boundary circle of the cap), and T^{00} is the local energy density. The CHM map sends the cap's causal development conformally to $\mathbb{R} \times H^2$, carrying the restricted vacuum to a thermal state at $\beta = 2\pi$ (in units of the curvature radius) [32, 33]. In the Lorentzian theory ξ generates a *boost-like* conformal flow. That K_{cont} has continuous spectrum and that the cap algebra is of type III are not statements of CHM or of Bisognano–Wichmann; they are standard facts of algebraic quantum field theory (under standard conditions local algebras are type III_1 factors [34], and for a type III_1 factor Connes' invariant $S(\mathcal{M})$, the intersection of the modular spectra, is the whole half-line $[0, \infty)$). The Bisognano–Wichmann 2π is a KMS parameter, not a periodicity of the real-time flow. Only the *Euclidean* continuation of this flow has discrete spectrum and a periodic flow: the Euclidean section is (conformally) S^3, and the continued flow is the rotation of S^3 that fixes a great circle (the Euclidean image of the entangling circle), closing at angle 2π; the KMS $\beta = 2\pi$ is exactly that Euclidean periodicity. The comparison below is with this Euclidean rotation generator, not with K_{cont} itself.

5.2 Framework wedge generator

The framework's posited wedge generator (Paper 42 Def 5.1) is

$$K_\alpha^W = \text{diag}(2m_j) \text{ on the wedge states},$$

a relabelling built from the polar-axis rotation generator $J_{\text{polar}} = 2J_z$ on the truncated Camporesi–Higuchi spectral triple (which itself anticommutes with the wedge reflection, so its compression to the wedge is zero). Its spectrum is

$$\text{spec}(K_\alpha^W) = \{2m_j : m_j = 1/2, 3/2, \ldots, (2n_{\max} - 1)/2\} = \{1, 3, 5, \ldots, 2n_{\max} - 1\},$$

integer-valued (the values of $2J_z$ on the wedge states), with multiplicity $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$. The posited modular flow $\sigma_t = \text{Ad}\, e^{itK_\alpha^W}$ satisfies $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$ bit-exactly at every tested $n_{\max}$ [12]. The odd-integer spectrum, the multiplicities, and the 2π closure are verified from the live BW construction in `tests/test_paper50_wedge_kms.py`.

Remark 5.1 (Normalisation of K_α^W and of ρ_W)**.** K_α^W carries the values $2m_j$ of $2J_z$, where J_z generates rotations by angle θ about the polar axis ($e^{-i\theta J_z}$); on the wedge it is the relabelling of [12] Def 5.1, not a restriction of $2J_z$. Three consequences. (i) The adjoint flow $\text{Ad}\, e^{itK_\alpha^W}$ already closes at $t = \pi$: differences of odd integers are even. The 2π closure above holds but is twice the minimal period; the unitary $e^{itK_\alpha^W}$ itself first returns to 1 at $t = 2\pi$ (it is -1 at $t = \pi$, the spinor sign). In the rotation-angle parameter $\theta = 2t$ the adjoint flow closes at $\theta = 2\pi$, which is the Euclidean periodicity matched in Observation 5.2. (ii) A state normalised as in Bisognano–Wichmann, with KMS parameter 2π in the rotation angle, would be $e^{-\pi K_\alpha^W}/Z$ (the same relabelling applied to $e^{-2\pi J_z}$), not $\rho_W = e^{-K_\alpha^W}/Z$; ρ_W is the unit-β Gibbs state of K_α^W (inverse temperature 2 in rotation-angle units). (iii) For $\rho_b = e^{-bK_\alpha^W}/Z$ one has $Z \simeq n_{\max}^2/(2\sinh b)$ and $\langle K_\alpha^W \rangle \to \coth b$, so $S(\rho_b) = 2\log n_{\max} + b\coth b - \log(2\sinh b) + O(1/n_{\max})$: the slope-2 headline is independent of b, while the constant $\coth 1 - \log(2\sinh 1)$ of (12) is the $b = 1$ value and is *not* a Bisognano–Wichmann-normalised quantity (at $b = \pi$ the constant would be $\pi\coth\pi - \log(2\sinh\pi) = 0.0136\ldots$).

Observation 5.2 (CHM analogy at finite cutoff)**.** ***[SYMBOLIC]*** *(the framework-side items) /* ***[OPEN]*** *(the identification). The framework's posited wedge generator K_α^W at finite $n_{\max}$ is a structural analog of the* Euclidean *rotation generator whose Wick rotation is the continuum CHM modular Hamiltonian K_{cont} of (13) — not of K_{cont} itself, which has continuous spectrum:*

- *K_α^W has odd-integer spectrum; the Euclidean rotation generator, in the same $2J$ normalisation, has discrete (integer) spectrum (continuum statement, prior art).*
- *Each generates a flow closing at rotation angle 2π (for K_α^W, carrying the values of $2J_z$, the adjoint flow closes at $t = \pi$; Remark 5.1); for the continuum this is the Euclidean periodicity that the Lorentzian modular flow of K_{cont} exhibits only as its KMS parameter $\beta = 2\pi$. The state ρ_W used in Sec. 4 is not normalised to that β.*
- *The framework's odd-integer spectrum $\{1, 3, \ldots, 2n_{\max} - 1\}$ corresponds to the half-integer m_j grading of the polar rotation Killing field on the (two-component, 3d) Dirac spinor sector.*

The integer spectrum and the 2π period are automatic for any integer-spectrum rotation generator [12], so they do not single out a modular Hamiltonian. What is not *established: that K_α^W is the modular Hamiltonian of a wedge-restricted Camporesi–Higuchi state (the open falsifier of [12]), and any convergence $K_\alpha^W \to K_{\text{cont}}$ as operators. The state-space Gromov–Hausdorff convergence of the spectral triples ([9], Fejér-kernel bound $(4/\pi)\log n_{\max}/n_{\max}$) is a statement about the triples, not about this operator.*

Basis of the analogy. The polar rotation Killing field of the round (Euclidean) S^3 — the Euclidean rotation generator, not the Lorentzian boost-like K_{cont} — rotates the spinor harmonics by their m_j eigenvalue, giving the integer $2m_j$ spectrum. The framework's wedge operator carries the same $2m_j$ values on the projected wedge sub-space $P_W \mathcal{H}_{\text{GV}}^{n_{\max}}$, restricted to the multiplicities allowed by the truncation, by the relabelling of Paper 42 Def 5.1 rather than by restricting J_{polar}.

6 Bulk-side scope

The three boundary-side observables (Sec. 3, Sec. 4, Sec. 5) re-derive a known continuum CFT$_3$-on-S^3 value from the continuum spectra (A), give an exact finite-cutoff degeneracy asymptotic (B), or parallel a continuum object (C, an analogy with the identification open). The corresponding *bulk-side* identifications — AdS$_4$ gravity, JLMS bulk-modular reconstruction, RT minimum surfaces, holographic entanglement entropy — are blocked at the framework's existing infrastructure.

Proposition 6.1 (Bulk side blocked)**.** *None of the bulk-side identifications corresponding to Sec. 3–5 are reachable from the framework's current infrastructure:*

1. *No AdS_4 / H^4 machinery exists in the framework's codebase (`geovac/`); a search for AdS, hyperbolic-4-space, or holographic primitives returns no matches.*
2. *Sprint RH-B (2026-04-17) [6] closed the natural candidate — Wick continuation $S^3 \to H^3$ — as a clean structural dead end: the continuation supplies no natural discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants.*
3. *Sprint L3e-P3 (2026-05-23) found that Paper 38's $4/\pi$ Fejér-moment constant does* not *transport to the non-compact Coulomb setting, suggesting that any discrete-to-bulk convergence statement would require a non-compact rate analog that is itself an open NCG-math problem.*

The structural reading of Prop. 6.1 is that the bulk-side identification falls into Class 1 (calibration-external) of the framework's external-input partition [40]: the framework evaluates the boundary-side quantities but does not autonomously generate the bulk-dual geometry. Importing AdS_4 infrastructure would be a major structural extension, beyond sprint scale and outside the present paper's scope.

7 Extension to S^5: structural refinement of the dual-basis decomposition

The closed-form Hurwitz evaluation of Sec. 3 extends to the continuum spectra on round S^5. The Camporesi–Higuchi spectra on S^5 are spectrally analogous to S^3: the conformally coupled scalar has eigenvalues $(n+3/2)(n+5/2)$ with multiplicity $(2n+4)(n+1)(n+2)(n+3)/24$ for $n = 0, 1, 2, \ldots$, and the Weyl Camporesi–Higuchi Dirac has $|\lambda_n| = n + 5/2$ with multiplicity $(n+1)(n+2)(n+3)(n+4)/12$. The same zeta-regularised evaluation of these untruncated spectra produces partition function values in a closed-form ring extending [24]'s S^3 ring by one $\zeta(5)/\pi^4$ generator.

7.1 Closed forms on S^5 (PSLQ-identified)

Theorem 7.1 (Conformally coupled scalar on S^5). ***[NUMERICAL, 200 dps + PSLQ; value prior art]*** *The zeta-derivative of the continuum conformal-scalar spectrum on round unit S^5 satisfies*

$$F_s^{S^5} := -\tfrac{1}{2}\,\zeta'_{\Delta_{conf}}(0)^{S^5} = -\frac{\log 2}{128} - \frac{\zeta(3)}{128\pi^2} + \frac{15\,\zeta(5)}{256\pi^4} \approx -5.74 \times 10^{-3}. \tag{14}$$

Proof sketch. The Hurwitz expansion

$$\zeta_{S^5}^{\mathrm{conf}}(s) = \frac{1}{12}\sum_{k\geq 0} g_k(s)[\zeta_R(2s+2k-4) - \zeta_R(2s+2k-2)] \tag{15}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ follows from the re-indexed multiplicity structure $v^2(v^2-1)/12$ at $v = n+2$. At $s = 0$, only $g_0(0) = 1$ survives, giving $\zeta_{S^5}^{\mathrm{conf}}(0) = 0$; the derivative

$$\zeta'_{S^5}(0) = \frac{1}{6}[\zeta'_R(-4) - \zeta'_R(-2)] + \frac{1}{12}\sum_{k\geq 1}\frac{\zeta_R(2k-4) - \zeta_R(2k-2)}{k\cdot 4^k} \tag{16}$$

converges geometrically in k. PSLQ at 200 dps and $k_{\max} = 100$ finds the integer relation $[32, -2, -2, 15]$ (applied to $4\,\zeta'_{S^5}(0)$), identifying

$$\zeta'_{S^5}(0) = \frac{\log 2}{64} + \frac{\zeta(3)}{64\pi^2} - \frac{15\,\zeta(5)}{128\pi^4}. \tag{17}$$

The standard identities $\zeta'_R(-2) = -\zeta(3)/(4\pi^2)$ and $\zeta'_R(-4) = +3\zeta(5)/(4\pi^4)$ deliver the $\zeta(5)/\pi^4$ contribution; the $\log 2$ piece arises, as on S^3, from resumming the Hurwitz-asymptotic tail (an earlier version attributed it to $\zeta'_R(0) = -\frac{1}{2}\log(2\pi)$, which would also bring in $\log\pi$; withdrawn 2026-09-29). □

Theorem 7.2 (Weyl Camporesi–Higuchi Dirac on S^5). *The zeta-derivative of the continuum Weyl Camporesi–Higuchi Dirac spectrum on round unit S^5 satisfies, as an exact symbolic identity,*

$$D'_{Dirac}(0)^{S^5} = -\frac{3\log 2}{128} - \frac{5\,\zeta(3)}{128\pi^2} - \frac{15\,\zeta(5)}{256\pi^4} \approx -0.02163. \tag{18}$$

Proof sketch. The Dirichlet series

$$D_{\text{Dirac}}(s)^{S^5} = \frac{1}{12}\left[\zeta_H(s-4,5/2) - \tfrac{5}{2}\zeta_H(s-2,5/2) + \tfrac{9}{16}\zeta_H(s,5/2)\right] \tag{19}$$

follows from re-indexing $u = n + 5/2$ in the multiplicity $(n+1)(n+2)(n+3)(n+4)/12 = (u^2 - 9/4)(u^2 - 1/4)/12$. The half-integer Hurwitz function $\zeta_H(s, 5/2)$ has closed form $(2^s - 1)\zeta_R(s) - 2^s - (2/3)^s$. Differentiating each shift symbolically at $s = 0$ and combining via the multiplicity coefficients $(1, -5/2, 9/16)$ delivers the claimed result. □

7.2 Log 3 cancellation theorem

Theorem 7.3 ($\log 3$ cancellation on S^5)**.** *The $(2/3)^s$ terms in $\zeta_H(s-a, 5/2)$ for $a \in \{0, 2, 4\}$ contribute, at $s = 0$, factors of $\log(2/3) = \log 2 - \log 3$ scaled by $(2/3)^{-a} = (3/2)^a$. The coefficient of $\log 3$ in $D'_{Dirac}(0)^{S^5}$ evaluates to*

$$\frac{1}{12}\left[\frac{81}{16} - \frac{5}{2}\cdot\frac{9}{4} + \frac{9}{16}\cdot 1\right] = \frac{1}{12}\left[\frac{81 - 90 + 9}{16}\right] = 0 \tag{20}$$

exactly.

Remark 7.4. The structural combination $(1, -5/2, 9/16)$ encoded in the multiplicity polynomial $(u^2-9/4)(u^2-1/4) = u^4-(5/2)u^2+9/16$ aligns the three shift operators $\{\zeta_H(s, 5/2), \zeta_H(s-2, 5/2), \zeta_H(s-4, 5/2)\}$ such that all $\log 3$ contributions cancel exactly. The S^5 values stay in the M-engine ring $\{\log 2, \zeta(3)/\pi^2, \zeta(5)/\pi^4\}$, even though the naive Hurwitz expansion suggests a contribution from $\log 3$ — a level-3 cyclotomic period, which would lie outside M3's level-≤ 4 ring $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ in the stratification of [2]. The same mechanism extends the ring statement from S^3 to S^5: $\log 2$ in M3's ring, and $\zeta(3)/\pi^2$, $\zeta(5)/\pi^4$ as M1×M3 cross-products.

7.3 Dual-basis projection: structural-specificity to S^3

Theorem 3.3 of Sec. 3 establishes that on S^3, the (scalar, Dirac) pair admits a dual-basis projection onto its two sectors: $F_D + 2F_s = \log(2)/2$ (the $\log 2$ sector only), $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ (the $\zeta(3)/\pi^2$ sector only). On S^5, the analogous statement *fails*.

Proposition 7.5 (Dual-basis non-extension to S^5)**.** *The Paper 50 Theorem 3.3 dual-basis projection does not extend to S^5. The M-engine ring on S^5 is three-dimensional ($\log 2$, $\zeta(3)/\pi^2$, $\zeta(5)/\pi^4$), but the (scalar, Dirac) plane is two-dimensional. The system $aF_s^{S^5} + bF_D^{S^5} = c \cdot$ (single ring generator) is over-determined: imposing isolation of one generator requires three linear conditions on two parameters (a, b), which has no nontrivial solution.*

Explicitly, attempting to isolate $\log 2$:

$$\zeta(3)/\pi^2 \text{ coefficient} = 0 \implies a + 5b = 0 \implies a = -5b, \tag{21}$$

$$\zeta(5)/\pi^4 \text{ coefficient} = 0 \implies 15a - 15b = 0 \implies a = b, \tag{22}$$

which are inconsistent unless $b = 0$.

Remark 7.6 (Continuation of the F-coefficient ladder through S^{11}; dual-basis non-extension)**.** Two statements must be distinguished here: the *single-species* in-ring closure of F (which continues to all odd d tested), and the *dual-basis projection* of Theorem 3.3 (which is special to S^3). The structural-specificity of Theorem 3.3 concerns only the second.

Single-species in-ring closure (continues through S^{11}). For each free species individually, $F = -\frac{1}{2}\zeta'_\Delta(0)$ on round S^d (d odd) lies in the ring

$$\mathcal{R}_d = \{\log 2\} \cup \{\zeta(2k+1)/\pi^{2k} : k = 1, \ldots, (d-1)/2\}$$

of dimension $(d-1)/2+1$ (top atom $\zeta(d)/\pi^{d-1}$). This is verified (symbolically or by PSLQ identification) at S^3, S^5 (Theorems 7.1, 7.2), and — by direct high-precision computation (Door 1 graduation, 2026-06-03; 260 dps, frozen basis, reconstruction error $\lesssim 10^{-261}$, decoy-controlled) — at S^7, S^9, and S^{11}. The conformally coupled scalar on S^{11}, for instance, is

$$\begin{aligned}\zeta'_{\mathrm{sc},S^{11}}(0) = &-\tfrac{7}{131072}\log 2 - \tfrac{3897}{45875200}\tfrac{\zeta(3)}{\pi^2} - \tfrac{485}{8257536}\tfrac{\zeta(5)}{\pi^4} \\ &+ \tfrac{609}{1310720}\tfrac{\zeta(7)}{\pi^6} + \tfrac{425}{262144}\tfrac{\zeta(9)}{\pi^8} + \tfrac{1023}{524288}\tfrac{\zeta(11)}{\pi^{10}},\end{aligned} \tag{23}$$

and the free Weyl Camporesi–Higuchi Dirac value is symbolic-exact in the same ring. A decoy test assigns Catalan G, $\log 3$, and unscaled $\zeta(3)$ coefficient exactly zero, confirming a clean signature: $\log 2$ (M3's ring, level 2) and the M1×M3 cross-products $\zeta(2k+1)/\pi^{2k}$.

Erratum. An earlier version of this remark reported S^7 as a structural non-match ("no PSLQ relation in the tested bases"). That was a *false negative* from an under-resolved search. The genuine S^7 scalar relation

$$\zeta'_{\mathrm{sc},S^7}(0) = -\tfrac{1}{512}\log 2 - \tfrac{41}{15360}\tfrac{\zeta(3)}{\pi^2} + \tfrac{5}{1024}\tfrac{\zeta(5)}{\pi^4} + \tfrac{63}{2048}\tfrac{\zeta(7)}{\pi^6}$$

has integer coefficients of order 10^4, but locking it requires the value to $\gtrsim 200$ digits; a 30-digit value run against maxcoeff $= 10^{16}$ sits below the resolution threshold and returns spurious non-detection. Verified independently at 200 dps (reconstruction error $\approx 2.4\times 10^{-202}$) and reproduced by the Door 1 driver at 260 dps. The ladder therefore *closes* (these are closed-form evaluations of continuum spectral zeta functions): every odd rung $S^3,\ldots,S^{11}$ has a rational-coefficient closed form in $\mathcal{R}_d$, with no transcendental outside $\mathcal{R}_d$.

Dual-basis projection (special to S^3, unaffected). Isolating a single ring generator using only the (scalar, Dirac) pair requires $\dim \mathcal{R}_d = 2$, which holds at S^3 alone. For $d \geq 5$ the projection is over-determined (Prop. 7.5); the related cross-species ladder recursion $\mathrm{scalar} - m\cdot\mathrm{Dirac} = c_d\cdot\mathrm{Dirac}_{S^{d-2}}$ likewise closes at S^5 only, with the top-atom ratio $\mathrm{scalar}_{\mathrm{top}}/\mathrm{Dirac}_{\mathrm{top}} = 2^{-(d-5)/2}$ hitting the required value ($m=1$) at $d=5$ alone (Door 1 graduation). Thus the F-coefficients are in-ring at every odd d, but the two-species dual basis that *spans* the ring is special to S^3. Supplying the missing higher-spin free species — $(d-1)/2-1$ of them at S^d — to span $\mathcal{R}_d$ remains the natural continuation, and is the actual structural-specificity content of Theorem 3.3.

7.4 Implementation effort and continuation pattern

The S^5 extension uses the same closed Hurwitz machinery on the continuum spectra (no truncation enters). The `debug/ads_track_a_s5_{scalar, dirac}_partition_function.py` scripts implement the Hurwitz expansions and PSLQ verifications; `debug/ads_track_a_s5_extension_memo.md` documents the findings in full.

8 Synthesis and open questions

8.1 Unified picture

Three structurally distinct calculational operations appear here, the first on the continuum spectra and the other two on the truncated triple with the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ built on it:

Operation	Continuum analog	Ring
$-\frac{1}{2}\zeta'_\Delta(0)$, continuum (Theorem 3.1)	F-theorem coefficient (KPS, re-derived)	M3-ring + M1×M3
$-\mathrm{Tr}[\rho_W \log \rho_W]$ (Prop. 4.1)	none claimed (lowest-shell degeneracy of K_α^W)	ring-free
$\mathrm{spec}(K_\alpha^W)$ (Observation 5.2)	Euclidean CHM rotation generator (analogy)	ring-free

The first row is evaluated on the continuum spectra directly and does not involve the truncation. The truncated triples on which the second and third rows are computed converge as $n_{\max} \to \infty$ in the van Suijlekom state-space Gromov–Hausdorff distance to the continuum triple per [9] (with the signature-agnostic product-carrier rate of [14] on the temporal factor); that convergence does not produce the F values of the first row. The second row is a finite-cutoff degeneracy count with no continuum identification claimed; the third row is an analogy with the Euclidean rotation generator only, and no operator convergence $K_\alpha^W \to K_{\text{cont}}$ is claimed (Observation 5.2).

8.2 Open questions

1. **Other free-field CFTs.** The scalar and Dirac cases exhibit the orthogonal $\log 2$ / $\zeta(3)/\pi^2$ decomposition of Theorem 3.3. Does the same projection structure extend to higher-spin partition functions on S^3, in particular the Maxwell ($p = 1$ form) and graviton ($p = 2$) cases? The Maxwell partition function on round S^3 is the natural next case; evaluating it by the same closed Hurwitz machinery (and testing whether the dual-basis decomposition extends) is a natural follow-up.

2. **Squashed-S^3 generalization.** The KPS values above are for the round (unit) S^3. The squashed-S^3 partition functions of supersymmetric CFTs [26] use deformed half-integer Hurwitz machinery that may not reduce to the framework's MR-B Jacobi-ϑ_2 closed form. Whether the framework can match the squashed case is open.

3. **State-space-GH-rate convergence statement.** The spectral triples converge at the state-space Gromov–Hausdorff level (Paper 38); Observation 5.2 does not inherit an operator-level rate from this (withdrawn 2026-09-28). Nor does Sec. 3 define or use a finite-cutoff free energy $F_{n_{\max}}$: its values are computed from the untruncated spectra. Whether a natural finite-cutoff free energy exists whose limit is F^{KPS}, with a rate $|F_{n_{\max}} - F^{\text{KPS}}| \leq C \cdot \gamma_{n_{\max}}$, would require a Mellin-domain version of the Berezin-reconstruction framework of Paper 38; this is open.

4. **Bulk-side identification (Prop. 6.1).** The JLMS bulk-modular Hamiltonian identification, the RT entanglement-entropy formula, and the AdS_4 dual geometry remain open. Importing AdS_4 / H^4 infrastructure is the named major structural extension that would reach these; it is beyond sprint scale.

8.3 Catalogue and open targets

The closed-form evaluations at S^3 and S^5 exhibit a pattern: zeta-regularised determinants of the continuum sphere spectra, evaluated in closed Hurwitz form, re-derive *independently-published* CFT-on-sphere free-energy values, and their transcendental signature decomposes via the master Mellin engine [2]. This pattern admits systematic extension. The following table catalogues the current status:

System	
Free scalar (conformal) on S^3	
Free Dirac (CH) on S^3	
Free scalar (conformal) on S^5	
Free Dirac (CH) on S^5	general-d formula of Giombi–Klebanov [23] (their F_f is per D
Free scalar (conformal) on S^7	
Free Maxwell on S^3	
Free Maxwell on S^5	
Free higher-rank tensor ($p \geq 2$) on S^d	a
Free (non-)minimal scalar (zero mode) on S^d	
Squashed S^3 partition function	

Holographic Weyl anomaly ↔ Connes–Chamseddine spectral action: both live in M2. A structural classification shared with the AdS/CFT literature that the framework obtains directly — without additional computation — is the M2-sector placement of the holographic Weyl anomaly and the Connes–Chamseddine spectral action.

Henningson and Skenderis [29] derived the Weyl anomaly of CFT_d on a manifold M via the bulk AdS_{d+1} gravitational action regularized at a finite radial cutoff. For even d, the result reproduces the boundary central charges $a_{\mathrm{CFT}}, c_{\mathrm{CFT}}$ as specific combinations of bulk gravitational coefficients. The bulk integral is computed via the Riemannian curvature data of AdS, evaluated through a heat-kernel expansion in the AdS-radius cutoff ϵ.

The Connes–Chamseddine spectral action [31] $\mathrm{Tr}\, f(D/\Lambda)$, on any almost-commutative spectral triple, has heat-kernel expansion coefficients given by Seeley–DeWitt coefficients $a_k(D^2)$. For the GeoVac spectral triple on round S^3, the Seeley–DeWitt coefficients of D^2_{CH} are exact, with no higher coefficients (Paper 28 [10], Sprint MR-B closed form, two-term-exact theorem); they carry an overall $\sqrt{\pi}$ from the heat-trace prefactor $\mathrm{Vol}(S^3)(4\pi t)^{-3/2}$, and once that standard normalisation is divided out they lie in the pure-Tate M2 ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ of the master Mellin engine (Paper 18 [2] §III.7). The spectral action therefore lives in the M2 sector.

Structural classification. Both the Henningson–Skenderis holographic Weyl anomaly and the Connes–Chamseddine spectral action sit in the master Mellin engine's M2 sub-mechanism (Seeley–DeWitt heat-kernel ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, after the standard $\sqrt{\pi}$ heat-trace normalisation is divided out). This is a shared heat-kernel *classification*, not a computed identity: both are heat-kernel-trace observables ($\mathcal{M}[\mathrm{Tr}\, e^{-tD^2}]$ at operator order $k = 2$), the HS quantity computed on the bulk-gravity side and the CC quantity on the finite spectral triple. The framework does not compute the HS holographic anomaly; the content is the taxonomic placement of both in the M2 ring. The framework's discrete-cutoff realization (Paper 38 [9] state-space GH convergence, with the Fejér-kernel bound $(4/\pi)\log n/n$ in the dual-Coxeter convention; $2/\pi$ on the unit S^3) is the operator-system-level home of the CC side.

This is the framework's most direct contact point with established holographic computations in the M2 sector and is read through the master Mellin engine's M2 slot (Paper 32 §VIII; the conditional case-exhaustion theorem covers chains of GeoVac projections, not this bulk computation), not a new computation. The substantive content is the explicit identification of HS's bulk computation as an M2-sector observable, placed in the same heat-kernel ring as the framework's spectral action on S^3 — a structural classification, not a reproduction of the holographic computation.

Relation to the TT-bar cut-off AdS picture. Hartman, Kruthoff, Shaghoulian, and Tajdini [30] showed that sphere partition functions of TT-bar-deformed large-N holographic CFT_d in $d \in \{2, 3, 4, 5, 6\}$ non-perturbatively match bulk computations of AdS_{d+1} at finite radial

cut-off. A parallel with the framework's spectral cutoff $n_{\max}$ is tempting, but the two cut-off structures are *categorically distinct*:

- **TT-bar (Hartman et al. 2019):** a *deformation of the action*. The TT-bar operator $\mathcal{O}_{T\bar{T}}(\mu)$ modifies the field theory's UV at finite μ; the deformed CFT has a different energy spectrum and a different action. The bulk dual is AdS at finite radial cutoff r_c related to μ via a (dimension-dependent) explicit formula.
- **GeoVac $n_{\max}$ (this paper):** a *truncation of the spectrum*, not the action. The framework's discrete spectral triple at $n_{\max}$ keeps the underlying free CFT and its action intact, but restricts the mode sum to $n \leq n_{\max}$. The F values of Sec. 3 (Theorems 3.1, 3.2) are computed from the untruncated spectrum, not as a limit of these truncations; the intermediate $n_{\max}$ values are not a TT-bar-deformed CFT but genuine spectral truncations of the original free CFT.

At the partition-function-value level, the two regularizations might still be related, but this paper constructs no finite-cutoff $Z_{n_{\max}}$ (a sharp mode truncation of $\log Z$ diverges with $n_{\max}$ and would need renormalisation). Whether a suitably renormalised $Z^{\mathrm{GeoVac}}_{n_{\max}}(R)$ equals $Z_{\text{TT-bar}}(\mu(n_{\max}, R), R)$ for some $\mu(n_{\max}, R)$ is an open question. A positive answer would identify the framework's spectral cutoff with a specific TT-bar-deformation strength; a negative answer would crystallize that the two cutoffs are structurally incompatible at the partition-function level.

Not holographic. The framework contains no bulk: its S^3 is the Fock-projection image of the Coulomb spectrum (Paper 7 [1]), not the conformal boundary or future-infinity boundary of any four-dimensional geometry, and there is no holographic dictionary mapping bulk fields to boundary operators. It is therefore not holographic, and it is neither an instance nor a refutation of AdS/CFT or of dS/CFT [41, 42] (in whose Euclidean form the CFT$_3$-on-S^3 partition function is the boundary datum). The F values of Sec. 3 are the standard zeta-regularised determinants of the continuum operators, evaluated here in closed Hurwitz form. Separately, Paper 38 [9] shows that the truncated triples converge to that continuum spectral triple in state-space Gromov–Hausdorff distance; that convergence is a statement about the triples and does not produce the F values. An earlier version of this paragraph proposed a third, non-holographic "correspondence" category with state-space GH convergence as its mechanism and credited the discrete triple with the boundary data; that reading is withdrawn (2026-09-29), since the F values never pass through the truncation.

Areas yet to be explored. Beyond the immediate catalogue targets, several adjacent directions extend the framework's reach into established physics:

- **Holographic central charges** (a- and c-anomalies in even d). Henningson-Skenderis [29] and follow-ups give closed forms for the holographic dual of free CFT central charges. The framework's spectral-action machinery (Paper 28 [10]) potentially provides a closed-form evaluation; the connection to Paper 24's Bargmann-Segal S^5 is unexplored.
- **Entanglement-entropy universal terms.** The continuum Casini-Huerta-Myers [32] entanglement entropy of a ball region has a universal subleading constant equal to $-F$. Connecting this to the framework's wedge KMS structure (Sec. 4) and to the master Mellin engine decomposition would strengthen the F-as-engine-signature reading.
- **Higher-rank Lie groups beyond** SU(2)**.** The Paper 40 unified GH-convergence theorem [11] extends the state-space GH convergence machinery to simple compact Lie groups (for general G conditionally on the per-group spin-window decomposition). A natural

catalogue extension is CFT-on-G partition functions for various G (heat-kernel zeta on SU(3), Sp(2), etc.) where the framework's Fejér-moment constant $4/\pi$ is reported universal (Paper 40; derived at rank 1) but the specific transcendental ring may differ.

- **Spectral-action coefficients.** The Connes-Chamseddine [31] spectral action $\mathrm{Tr}\, f(D/\Lambda)$ has heat-kernel expansion coefficients that classify the gravitational + matter content of an almost-commutative spectral triple. These coefficients are exactly the master Mellin engine M2 sector. The catalogue could include M-engine readings of standard spectral-action calculations on S^3, S^5, and higher.

- **Lorentzian-side observables.** The catalogue here is boundary-side Riemannian. The Lorentzian papers (Paper 42; Papers 46–49, now archived or descoped) posited a wedge modular Hamiltonian (its identification remains open) and studied Pythagorean orthogonality and twin-paradox-as-quantum-information structures. Cataloguing Lorentzian-side CFT observables (thermal partition functions on $S^3 \times \mathbb{R}_t$, Rindler-wedge entropies, etc.) is a parallel direction.

Each catalogue entry either (i) classifies independently published continuum values under the master Mellin engine, (ii) extends a structural cancellation theorem (like log 3 cancellation on S^5, Theorem 7.3), or (iii) identifies a wall where the simple ring fails (like the squashed-S^3 deformed-Hurwitz case, not in the MR-B Jacobi-ϑ_2 closed form). Each outcome is paper-worthy in its own right; the cumulative catalogue would constitute a structural classification of free-field CFT-on-sphere partition functions via the master Mellin engine.

8.4 Place in the GeoVac math.OA series

This paper is the twelfth math.OA standalone in the GeoVac series, following Papers 38, 39, 40, 42, 43 (archived 2026-09-28; its surviving content is Paper 42 § 11), 44, 45, 46, 47, 48, 49. The previous eleven established operator-algebraic legitimacy of the framework's discrete-spectral-triple truncation and the spectral-truncation convergence theory at the Riemannian and tensor-product levels and, for compact connected semisimple Lie groups (conditionally, for general G; the universal Fejér-moment constant for the simple ones), together with the norm-resolvent non-compact-carrier closure at the spectral level. The Lorentzian *propinquity* levels—strong-form Lorentzian (Paper 46) and both synthetic-Lorentzian bridges (Papers 48, 49)—are *descoped* by Paper 45's degeneracy theorem, consistent with this paper's introduction, and are not convergence results; Papers 46–49 are archived (`docs/retired_papers.md`). The present paper places that math.OA arc alongside a physics literature — the CFT-on-sphere partition function literature of [22, 24, 25, 27, 28] — with a $\log 2$ / $\zeta(3)/\pi^2$ (M3-ring / M1×M3) decomposition of independently published values (the master Mellin engine applied to the continuum KPS data), an independent closed-form re-derivation of those values from the continuum spectra (not from the truncation), an exact entropy asymptotic on the truncated triple (the lowest-shell degeneracy of the posited generator), and one analogy (the posited wedge generator against the Euclidean rotation generator of the CHM construction; the identification with a modular Hamiltonian is open).

Acknowledgments

This work was developed in the GeoVac project's AI-augmented agentic workflow with Anthropic's Claude.

References

[1] J. Loutey, *Paper 7: The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice*, GeoVac internal preprint, 2026.

[2] J. Loutey, *Paper 18: Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds*, GeoVac internal preprint, 2026.

[3] J. Loutey, *Paper 55: Periods of GeoVac: a cyclotomic mixed-Tate census of the master Mellin engine*, GeoVac internal preprint, 2026.

[4] J. Loutey, *Paper 22: Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems*, GeoVac internal preprint, 2026.

[5] J. Loutey, *Paper 23: Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice*, GeoVac internal preprint, 2026.

[6] J. Loutey, *Paper 29: Finite-Size Graph-RH for the GeoVac Hopf Graphs: Ihara Zeta, Bound-Crossing, and a Scope Boundary on Selberg-on-Hydrogen*, GeoVac internal preprint, 2026.

[7] J. Loutey, *Paper 32: The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage*, GeoVac internal preprint, 2026.

[8] J. Loutey, *Paper 35: Time as Projection: The Klein-Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split*, GeoVac internal preprint, 2026.

[9] J. Loutey, *Paper 38: State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3*, GeoVac internal preprint, 2026.

[10] J. Loutey, *Paper 28: QED on S^3: Spectral Geometry of Perturbative Transcendentals*, GeoVac internal preprint, 2026.

[11] J. Loutey, *Paper 40: State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric*, GeoVac internal preprint, 2026.

[12] J. Loutey, *Paper 42: Tomita–Takesaki modular structure on truncated $SU(2)$ spectral triples: the four-witness 2π period at finite cutoff*, GeoVac internal preprint, 2026.

[13] J. Loutey, *Paper 44: Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle*, GeoVac internal preprint, 2026.

[14] J. Loutey, *Paper 45: Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples: a degeneracy theorem*, GeoVac internal preprint, 2026.

[15] J. Loutey, *Paper 46: Strong-form Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples*, GeoVac internal preprint, 2026.

[16] J. Loutey, *Paper 47: Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on $S^3 \times \mathbb{R}_t$ and the two-rate hybrid limit*, GeoVac internal preprint, 2026.

[17] J. Loutey, *Paper 48: Krein-pointed proper quantum metric spaces and the bridge to Mondino–Sämann Lorentzian pre-length spaces, with application to the GeoVac hemispheric wedge*, GeoVac internal preprint, 2026.

[18] J. Loutey, *Paper 49: Operator-system Lorentzian pre-length spaces and the strong-form Krein–Mondino–Sämann bridge*, GeoVac internal preprint, 2026.

[19] V. Fock, *Zur Theorie des Wasserstoffatoms*, Z. Phys. **98** (1935), 145–154.

[20] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18.

[21] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067, arXiv:2004.14115.

[22] J. L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nucl. Phys. B **270** (1986), 186–204.

[23] S. Giombi and I. R. Klebanov, "Interpolating between a and F," *JHEP* **03** (2015) 117, arXiv:1409.1937.

[24] I. R. Klebanov, S. S. Pufu, and B. R. Safdi, *F-theorem without supersymmetry*, JHEP **10** (2011) 038, arXiv:1105.4598.

[25] D. L. Jafferis, I. R. Klebanov, S. S. Pufu, and B. R. Safdi, *Towards the F-theorem: $N = 2$ field theories on the three-sphere*, JHEP **06** (2011) 102, arXiv:1103.1181.

[26] N. Hama, K. Hosomichi, and S. Lee, *SUSY gauge theories on squashed three-spheres*, JHEP **05** (2011) 014, arXiv:1102.4716.

[27] M. Beccaria, X. Bekaert, and A. A. Tseytlin, *Partition function of free conformal higher spin theory*, JHEP **08** (2014) 113, arXiv:1406.3542.

[28] Y. Lei and S. van Leuven, *Modularity in $d > 2$ free conformal field theory*, arXiv:2406.01567, 2024.

[29] M. Henningson and K. Skenderis, *The holographic Weyl anomaly*, JHEP **07** (1998) 023, arXiv:hep-th/9806087.

[30] T. Hartman, J. Kruthoff, E. Shaghoulian, and A. Tajdini, *Holography at finite cutoff with a T^2 deformation*, JHEP **03** (2019) 004, arXiv:1807.11401.

[31] A. H. Chamseddine and A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997), 731–750.

[32] H. Casini, M. Huerta, and R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, arXiv:1102.0440.

[33] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.

[34] D. Buchholz, C. D'Antoni, and K. Fredenhagen, *The universal structure of local algebras*, Commun. Math. Phys. **111** (1987), 123–135.

[35] S. Ryu and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*, Phys. Rev. Lett. **96** (2006), 181602, arXiv:hep-th/0603001.

[36] D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, *Relative entropy equals bulk relative entropy*, JHEP **06** (2016) 004, arXiv:1512.06431.

[37] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, *Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence*, JHEP **06** (2015) 149, arXiv:1503.06237.

[38] GeoVac internal, *Sprint BH-Phase0 diagnostic memo*, `debug/bh_phase0_diagnostic_memo.md`, 2026-05-22.

[39] GeoVac internal, *Sprint TD Track 5 PSLQ-disjoint correlation entropy*, `debug/sprint_td_track5_memo.md`, 2026-05-09.

[40] GeoVac internal, *External-input three-class partition*, `memory/external_input_three_class_partition.m` 2026-05-25.

[41] A. Strominger, *The dS/CFT correspondence*, JHEP **10** (2001) 034, arXiv:hep-th/0106113.

[42] J. Maldacena, *Non-Gaussian features of primordial fluctuations in single field inflationary models*, JHEP **05** (2003) 013, arXiv:astro-ph/0210603.